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# NOTES ON A DEGENERATE PARABOLIC EQUATION

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Based on *Uniqueness and Root-Lipschitz  
Regularity for a Degenerate Heat Equation*

Author  
Wells Guan

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# 1 Introduction

## 1.1

**Definition 1.1** (Heat Equation and Adjoint equation). The heat equation

$$\partial_t u = \Delta u, \quad u \in \mathbb{R}^n \times \mathbb{R}$$

has a class of solution

$$\Gamma(x, t; \xi, \tau) = \frac{1}{(2\sqrt{\pi})^n} (t - \tau)^{-n/2} \exp\left(-\frac{|x - \xi|^2}{4(t - \tau)}\right), \quad (\xi, \tau) \in \mathbb{R}^n \times \mathbb{R}$$

where for any continuous function  $f(x) \lesssim \exp(h|x|^2)$  for some  $h > 0$ , then the integral

$$u(x, t) = \int \Gamma(x, t; \xi, 0) f(\xi) d\xi$$

exists for  $0 < t \leq T$  if  $T < 1/4h$  and  $u(x, t) \rightarrow f(x), t \rightarrow 0$ , and  $\Gamma$  is called a fundamental solution of the heat equation; and  $u$  is a solution of the Cauchy problem, i.e. the problem of finding a solution to the heat equation for  $0 < t \leq T$  with the initial condition  $u(x, 0) = f(x)$ .

The adjoint equation is defined as

$$\partial_t v = -\Delta v, \quad v \in \mathbb{R}^n \times \mathbb{R}$$

and in this case  $\Gamma(x, t; \xi, \tau)$  is a solution for the adjoint equation as a function of  $(\xi, \tau)$  with fixed  $(x, t)$ .

**Definition 1.2** (Holder Space). A function  $f(x)$  defined on a bounded closed set  $S$  of  $\mathbb{R}^n$  is said to be Holder continuous of exponent  $\alpha$  ( $0 < \alpha < 1$ ) in  $S$  if there exists a constant  $A$  such that

$$|f(x) - f(y)| \leq A|x - y|^\alpha$$

for all  $x, y$  in  $S$ . The smallest  $A$  for the inequality is called the Holder coefficient.

If  $S$  is an unbounded set whose intersection with every bounded closed set  $V$  is closed, then  $f(x)$  is said to be Holder continuous of exponent  $\alpha$  in  $S$  if for every bounded closed set  $V$  the equality holds in  $S \cap V$  with some  $A$  which may depend on  $V$ . If  $A$  can be taken independently of  $V$ , then we say that  $f(x)$  is uniformly Holder continuous.

If  $S$  is an open set then  $f(x)$  is said to be locally Holder continuous (exponent  $\alpha$ ) in  $S$  if the inequality holds in every bounded closed set  $V \subset S$ , with  $A$  which may depend on  $V$ . If  $A$  is independent of  $V$  then  $f(x)$  is uniformly Holder continuous (exponent  $\alpha$ ).

If  $f$  depends on a parameter  $\lambda$ , i.e.  $f = f(x, \lambda)$ , and if the Holder coefficient is independent of  $\lambda$ , then we say that  $f(x, \lambda)$  is Holder continuous in  $x$ , uniformly with respect to  $\lambda$ .

If in the inequality  $\alpha = 1$  then we say that  $f(x)$  is Lipschitz continuous.

**Definition 1.3** (Parabolic Equation). Consider the differential equation

$$Lu \equiv \sum_{i,j=1}^n a_{ij}(x, t) \partial_{x_i} \partial_{x_j} u + \sum_{i=1}^n b_i(x, t) \partial_{x_i} u + c(x, t)u - \partial_t u = 0$$

where the coefficients  $a_{ij}, b_i, c$  are defined in a cylinder

$$\Omega \equiv \overline{D} \times [T_0, T_1] \equiv \{(x, t); x \in \overline{D}, T_0 \leq t \leq T_1\}$$

$\overline{D}$  is the closure of a bounded domain  $D \subset \mathbb{R}^n$ . We always take  $a_{ij}(x, t)$  to be a symmetric matrix. If the matrix is positive definite, then we say that the operator  $L$  is of parabolic type at the point  $(x, t)$ . If  $L$  is parabolic at all the points of  $\Omega$  then we say that  $L$  is parabolic in  $\Omega$ . If there exist positive constants  $\bar{\lambda}_0, \bar{\lambda}_1$  such that, for any real vector  $\xi$ ,

$$\bar{\lambda}_0 |\xi|^2 \leq \sum_{i,j=1}^n a_{ij}(x, t) \xi_i \xi_j \leq \bar{\lambda}_1 |\xi|^2$$

for all  $(x, t) \in \Omega$  then we say that  $L$  is uniformly parabolic in  $\Omega$ .

**Remark 1.1.** Throughout this chapter we shall assume:

- (A1)  $L$  is parabolic in  $\Omega$ ;
- (A2) the coefficients of  $L$  are continuous functions in  $\Omega$  and, in addition, for all  $(x, t) \in \Omega, (x_0, t_0) \in \Omega$ , - - - (1.4)  $3(kh, t) \text{ id eg} \rangle < L(|a; a, -, -Cr0, sdeg| \ll + \hat{\text{id eg}} |deg/2)$ ,  
- -

|       |            |                             |       |
|-------|------------|-----------------------------|-------|
| (1.5) | $ 'Dzh, 0$ | $biCc0, 01 < A \setminus x$ | $xt,$ |
|-------|------------|-----------------------------|-------|

- -

|       |                                 |                               |              |
|-------|---------------------------------|-------------------------------|--------------|
| (1-6) | $ c(z, i)$                      | $c(rrn, i)   < A \setminus x$ | $rrdegK$     |
| Since | the $ats(kh, t)$ are continuous | functions in the bounded      | set $O, (A^$ |

implies that  $L$  is also uniformly parabolic in  $O$ , i.e., (1.3) holds for some positive constants  $l_0, l_1$  independent of  $(x, t) \in O$ .

**Definition.** We say that  $i - u(x, t)$  is a solution of  $Lu = 0$  in some

|                               |        |                                                   |
|-------------------------------|--------|---------------------------------------------------|
| rpgion                        | $D$    | if all the derivatives of $i$ which occur in $Lu$ |
| $\hookrightarrow$             |        | (i.e., $du/dx_{it}$                               |
| $0h_i/dx - i \, dx_j, di/dt)$ | are    | continuous                                        |
| functions                     | in $D$ | and $Lu(x, t) = 0$ at                             |

each point  $(x, i)$  of  $D$ . A similar definition holds for any differential  $L$ . operator

**Definition.** A

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| fundamental solution of $Lu \equiv 0$ (in $O$ ) is a function                                                  |
| $G(gg, i; PS, t)$ defined for all $(rr, 0 \leq PS \leq 0, (PS, 0 \leq PS \leq 0, i > t)$ , which satisfies the |

following conditions:

- |                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------|
| (i) for fixed $(PS, t) \in O$ it satisfies, as a function of $(z, i) \in (kh \leq Z)$ ,<br>$t < t < T_i$ ) the equation $Lu = 0$ ; |
| (ii) for every continuous function $f(x)$ in $\sim D$ , if $rr \in D$ then                                                         |

$$(1.7) \quad \lim_{t \rightarrow \infty} \int_D G(a, t; f, t) dx = f(x).$$

– Here  $df = df_i \cdot dx_i$ . The domain  $D$  is always assumed to be Lebesgue measurable; since this is an obvious assumption, we shall usually not mention it in what follows.

Sections 2-4 are devoted to the construction of a fundamental solution. In Sec 5 we shall study some of its properties. The results of Secs. 2-5 are extended in Sec. 6 to the case of unbounded domains  $D$ .

$$\begin{array}{ccccccc} 4 & & \text{FUNDAMENTAL} & & \text{SOLUTIONS} & & \text{AND} & \text{TPK} \\ \hookrightarrow & \text{CAUCHY} & \text{I'ltOULKM} & & (\text{II}\backslash 1\backslash L & & & \end{array}$$

## 2 The Parametrix Method

$$\text{Let } (a_i)(x, t) \text{ be the inverse matrix to } (a_j)(x, t) \text{ and} \\ \hookrightarrow \text{set}$$

$$= S - \sim$$

$$(2.1) \quad \text{un}''(z, 0) = a_H y, \quad s(kh) \\ \hookrightarrow \text{Sh}^{-1} \quad f(c)$$

$$g_{,j} = 1$$

$$\begin{array}{l} \text{where } u = (u_0, \dots, u_p). \text{ From (1-3), (1.4) it follows that} \\ - < - \\ (2.2) \quad l_0 |z| < \text{No. } (kh, 0 \\ \hookrightarrow lkh \backslash kh \quad x/2, \\ - \\ (2.3) \quad |a''(kh, t)| < L' \backslash kh \\ \hookrightarrow khdeg \backslash a + \quad \backslash t \quad tdeg \backslash a/2) \end{array}$$

where  $l_0, l_1, L'$  are positive constants depending only on  $l_0, l_1, L$ . We introduce the functions, for  $t > r$ ,

$$(2.4) \quad wv''\{x, i; x, r\} = \{i - r\}^{-1/2} \exp \\ \hookrightarrow [-\wedge^j$$

$$(2.5) \quad Z\{x, t; x, r\} = G(x, t)u^*(\cdot, i; x, g),$$

$$\begin{array}{l} \text{where} \\ = \end{array}$$

(2.6)  $C(x, t) = (2VV)^{-1} [\det (a'(zh, \rightarrow 0))]^{1/2}$   
 For each fixed  $(x, t)$  the function  $Z(x, t; PS, r)$   
 $\rightarrow$  satisfies the equation

with constant coefficients

$\rightarrow 0 < r < \infty$ ,  $n = 2$ ,  $\chi = \chi(x, t, r)$   
 $n = 0$  -  $\chi(x, t, r) = \chi(x, t, r)$   
 (2.7)  $L_0 u(x, t) = S(x, t, r)$   
 $\rightarrow$  -  $0$ .

$= \chi(x, t, r)$

It also follows from (1.7) that (1.7) is also satisfied for  $G = Z$ .

**Theorem 1 below.**

Thus,  $Z(x, t; PS, t)$  is a fundamental solution of  $L_0 u = 0$ . In order to construct a fundamental solution for  $L u = 0$  we look upon  $L$  as a "first

approximation" to  $L$  and we view  $Z$  as a "principal part" of the fundamental solution  $G$  of  $L u = 0$ . We then try to find  $G$  in the form

(2.8)  $G(x, t; x, t) = Z(x, t; x, t) + \chi(x, t; n, s) \Phi(x, t; s, x, g)$   
 $\rightarrow$   $0, r/s$

where  $\Phi$  is to be determined by the condition that  $G$  satisfies the equation  $L u = 0$ . This procedure is called the parametrix method (of K. K. Levi).  $Z$  is called the parametrix. In Chap. I we shall use this method also to construct fundamental solutions for systems of parabolic equations of any order.

**Theorem 1.** Let  $f(x, t)$  be a continuous function in  $O$ . Then

(2.9)  $J(x, t, r)$  is  $Z(x, t; x, r)/(x, t) dt$

is

is a continuous function in  $(x, t, r)$ ,  $0 < t < \infty$ ,  $I \setminus \{0\}$  and

(2.10)  $J(x, t) = \int_0^t Z(x, t; x, r) dr$

$t-E$

uniformly with respect to  $(x, t)$ ,  $\forall x \in S, T_0 < t < T$ , where  $S$  is any closed

subset of  $D$ .

*Proof.* Consider first the case where  $f$  and the  $a_i$  are all constants.

The linear substitution  $f = P(x)$  reduces  $S$  into  $S'$  provided  $P^*P = I$ , where  $P^*$  is the transpose of the matrix  $P$ .

$(a_i)$ , where

Denoting by  $D^*$  the image of  $D$  by this substitution and noting that

$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

we get

(2.11)  $J(x, y) = \frac{1}{2} \int_0^1 \dots$

$+ L$

where  $J_1$  is the part of the integral taken over a ball  $|f| < b$  with radius  $R$  sufficiently small (so that the ball is contained in  $D^*$ ), and  $J_2$  is the

complementary part. Introducing polar coordinates, we get

$J_1 = \int_0^1 \dots$

if

$(2W)^n \dots$

where  $2\pi \omega_2 =$

(1) is the surface area of the unit hypersphere. It follows that

|                                                                     |                                                 |
|---------------------------------------------------------------------|-------------------------------------------------|
| (2.13)                                                              | $\lim_{j \rightarrow \infty} J_j = \frac{1}{2}$ |
| $t \rightarrow t$                                                   |                                                 |
| Since $R$ is fixed, the integrand                                   | of $J_j$ tends to                               |
| $\rightarrow 0$ as $t \rightarrow \infty$ . Hence $J \rightarrow 0$ |                                                 |
| as $t \rightarrow i$ . Combining this remark                        | with                                            |
| $\rightarrow$ (2.13) it follows, by (2.11), that                    |                                                 |
| $J(x, t, t) \rightarrow \frac{1}{2}$ as $t \rightarrow t$ .         |                                                 |

Consider now the general case where  $\beta$  and the  $a_j$  are not constants. Writing

$$J(x, U, t) = \int_{\mathcal{C}(x, t)} \mathcal{L}(x, U, t) dt$$

fD

$$\begin{aligned} &+ L^* \cdot \\ &T) \quad GP_i, \quad t) < < *'(*, i; x, t) \quad S(kh, i) i i^{-(kh, i; x, t)}] < PS \\ (2.14) & \quad \quad \quad /d \end{aligned}$$

+

$$C(\{, r) < < *'(*, i; i, t) \quad / (x, t) \quad / (kh, t)] < 2PS$$

//;

$$= \frac{1}{i} + \frac{1}{L} + \frac{1}{L},$$

we may treat the integral of  $\llbracket L \rrbracket$  as  $J$  in the previous case of constant coefficients. Hence,

$$(2.15) \quad \lim Jx = f(x, t).$$

As for J2, -

|     |                |                  |
|-----|----------------|------------------|
| No. | 'T(x, t; x, t) | w*-l(x, l; x, t) |
|-----|----------------|------------------|

 $\sim$ 

$$\langle n^* \backslash x - i \backslash i$$

—

$$r) - (1 + 2V_{2m}x \quad \backslash a * > \& t)$$

—

$a \rightarrow (kh, 01 \text{ ekhrG}^- \quad \quad \quad -- \quad \quad *L-$

$w L 4(t t; j$

Writing the integrand of the integral of  $J_z$  in the form

$/-$   
 $No. 1*, \quad t) \quad \quad \quad C(x, t)]w*(x, \quad \quad \quad t; x, t)$

$-$

$+ \quad \quad C(z, t)[wt*(x, t; PS, r) \quad \quad \quad w?*(x, t; x, r)]$   
 and using the previous inequality we find that for any  $\epsilon > 0$  there  
 $\hookrightarrow$  exist

$R$  and  $d$  sufficiently small such that the expression  $I$  is bounded by

$-$   
 $(2.16) \quad \quad \quad e(t \quad \quad \quad t) \rightarrow */[1 \quad \quad \quad \backslash x \quad \quad \quad x|2(i \quad \quad \quad t)-1] \exp$   
 $[-14deg[*G*(2]$

if  $|x - x| < R$  and  $t$   
 $-- t < d.$

Dividing the integral of  $J_i$  into two parts,  $-$

say  $I \backslash + 12$  where for  $/1$  the integration is taken over  $\backslash x \quad \quad \quad x \backslash < R$ , and  
 estimating  $/1$  by using the bound (2.16) for its integrand and then using  
 as in (2.12), and substituting  $s = \quad \quad \quad --$   
 polar coordinates,  $r^2/(t \quad \quad \quad t)$ , we find  
 that  $|/i| < C\epsilon$  if  $t - t < d$ , where  $S$  is a constant independent of  $\epsilon$ . Now,  
 if  $R$  is fixed and  $r \rightarrow i$  then  $I_2 \rightarrow 0$  since its integrand tends to zero. It

follows that  $|J_2| < C't$  if  $t$  is sufficiently close to  $t$ , where  $S$  is a constant independent of  $\epsilon, t$ .

Hence

$\lim/5 = 0.$

(2.17)  
 $t \rightarrow <$

$J_3$  can be treated in a similar way. We break it into a sum  $J_{31} + J_{32}$   
 where the integration in  $J_{31}$  is taken over a ball  $|x - x| < R$ . Since  $f(x, t)$   
 is a continuous function, for any  $\epsilon > 0$  the integrand of  $J_{31}$  is bounded

by

<<- '>-'-'

$t-s^a$

provided  $b$  and  $t-t$  are sufficiently small. Introducing polar

as in (2.12), and substituting  $a = r^2/(t -$   
coordinates,  
 $\hookrightarrow$   $t$ ) we find that  $|J_3| < Ce,$   
 $S$  being a constant independent of  $e$ . For  $R$   
 $\hookrightarrow$  fixed,  $J_3 \rightarrow 0$  as  $t \rightarrow i$ . We  
 thus obtain:  $J_3 \rightarrow 0$  if  $r \rightarrow t$ . Combining  
 $\hookrightarrow$  this with (2.17), (2.15) and  
 recalling (2.14), we get (2.10).

The assertion concerning the uniform convergence follows from the previous proof.

### 3 Volume Potentials

Given a function  $f(x, t)$  in  $O$  we consider the function

$$(3.1) \quad V(x, t) = \int_0^t Z(x, t; x, t) / (x, t) \, dx \, dr$$

$\int \ln fD$

and call it the volume potential of  $f$  (with respect to the parametrix  $Z$ ).

We shall study in this section some differentiability properties of  $V$ . These will be used in the construction of a fundamental solution in the following section. Note that the volume potential is an improper integral, the integrand

having a singularity at  $PS = kh, t = t$ . The singularity, however,  
 $\hookrightarrow$  is in-

tegrable. Indeed, writing  $wy^7$  in the form

$$(3.2) \quad uR^{**}(kh, 1; x, t) \\ G \, LU.t \, "I(7I/2)-m \, G \, Lu, t \, - \backslash$$

—

(i, ), -, , (\*>>), -, /\* ekhr  
 [--j [-Shch^GT)}  
 where W-T = No. \*t(kh, x), we get

$$(3.3) \quad |u^m(z, t; x, T)| \leq \int_0^t \int_S |f(s, y)| dy ds + \int_0^t \int_S |g(s, y)| dy ds$$

and the right-hand side is integrable. Before proceeding to the actual study of  $V$ , we give a useful elementary lemma.

**Lemma 1.** Let  $f(x, y)$  be a continuous function of  $(x, y)$  when  $x, y$  vary in a compact domain  $S$  of  $\mathbb{R}^m$  and  $h \in \mathbb{R}^m$ , and let  $L > 0$

$$\int_S |f(x, y)| dy \leq L \text{ for all } x \in \mathbb{R}^m,$$

uniformly with respect to  $h$  in  $S$ , where  $S(x, e)$  is the intersection of  $S$  with the ball with center  $h$  and radius  $e$ . Then, for any bounded measurable

$$g(y) \text{ in } S, \text{ the (improper)}$$

$$\text{function integral} \\ =$$

$$\int_S f(x, y) g(y) dy$$

is a continuous function in  $S$ .

*Proof.* Given  $\epsilon > 0$  choose  $\delta$  such that

$$\int_S |f(x, y) - f(x', y)| dy < \epsilon \text{ for all } x, x' \in S, |x - x'| < \delta. \text{ Since } f(x, y) \text{ is a uniformly continuous (and hence a bounded) function of } (x, y) \text{ when } (x, y) \in S, \text{ we have } \delta >$$

$\delta/2$ , there

exists a  $\delta_2 < \delta/4$  such that

$$(3.4) \quad \int_S |f(x, y) - f(x', y)| dy < \epsilon$$

/

for all  $h \in S, z \in S$ . Denoting the complement of  $S(z, \delta_2)$  by  $S^c(z, \delta_2)$

we obtain, from (3.1),  
 $\sim < 2\epsilon$ .

$$(3.5) \quad \int_{\partial D} f(x, y) g(y) dy = \int_{\partial D} f(x, y) g(y) dy$$

for

Using the uniform continuity of  $f(x, y)$  for  $x \in S, y \in S$ ,  $|x - y| < \delta/2$

we have

$$< \epsilon$$

$$|f(x, y) - f(x', y)| \leq \epsilon$$

—

provided  $|x - x'| < \delta$  for some  $\delta$  sufficiently small ( $\delta < \delta/2$ ). Hence,  $|f(x, y) - f(x', y)| \leq \epsilon$  together with (3.5) (for  $z = x, z' = x', x \in S, x' \in S$ ) we get

$$|f(x, y) - f(x', y)| \leq 3\epsilon \int_{\partial D} dy$$

$$|f(x, y) - f(x', y)| \leq 3\epsilon$$

Combining this with (3.4) (for  $z = x, z' = x'$  and  $|x - x'| < \delta$ ) we derive the

—

inequality  $|f(x, y) - f(x', y)| \leq \epsilon$ , and the proof is completed.

define  $Z(x, t; x, t) = 0$  for  $t < t_0$  then

If we can apply Lemma 1 to the volume potential, and thus conclude:

**Theorem 2.** //  $f(x, t)$  is a bounded measurable function in  $O$  then the volume potential  $G(x, t)$  is a continuous function in  $O$ .

We next prove:

**Theorem 3.** //  $f(x, t)$  is a continuous function in  $O$  then  $V(x, t)$  has

first continuous derivatives with respect to  $x$  for  $x \in D, t_0 < t < t_1$  and

$$(36)$$

$$b^u_{-} | b_{AZ}(G>I; iitNo. \ t) (, \{ (1t.$$

*Proof.* Writing  $\text{dwy}'T(x, t; x, t)/\text{thkh}$ ,; analogously to (3.2) we find that

$$\frac{a}{w^{\wedge}(x, t; x, t)} \frac{\partial}{\partial x} \left( \frac{1}{(i t U \backslash k h x | e + i - 2 M \backslash 2 < m I} \right) \frac{1}{\text{const.}}$$

Thus, the singularity of the integrand in (3.6) is integrable. If we define

$dZ(x, \dots) = 0$  for  $t < t$  then we can apply Lemma  
 $\hookrightarrow$  I and thus  
 $/;PS, r)/dXi$

conclude that the integral of (3.0) is a continuous function. It remains to verify (3.0). Set

$$(3.8) \quad J\{x, t, r\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \delta(x - x_0) \delta(t - t_0) \delta(r - r_0) \dots \delta(x_n - x_{n0}) \delta(t_n - t_{n0}) \delta(r_n - r_{n0}) dx_0 dt_0 dr_0 \dots dx_n dt_n dr_n$$

Jd Z[x, t; x, t)No. , Then =

$$(3.9) \quad V(x, t) = \int_0^t J(x, t, r) \, dr.$$

By Theorem 1, Sec. 2,  $J(r, t, r)$  is a continuous  
 $\hookrightarrow$  function in  $(kh, I, t)$   
 when  $kh(z \in D, T_0 < t < T_1)$ , provided we define  
 $= \lim J(x, t, t)$ .  
 (3.10)  $J(x, t, t)$

t-E Clearly, if  $t > t$

$$dJ(x, t, t) = [d n \quad (3.11) \quad dX_j \quad /,,teZ(*,<<fer)rti,T)<it.$$

Using (3.7) it follows that

$$\frac{dJ(x, t, t)}{dt} = \text{const.} \quad (3.12)$$

Consequently, the improper integral

$$(3.13) \quad \int_1^{\infty} \frac{V(x, t)}{x^{s+1}} dx$$

is absolutely uniformly convergent.

Let  $x_h = (p, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n)$  and  
 $\hookrightarrow$  consider

$$(3.14) \quad \frac{V(x \setminus t)}{h} = \frac{V(x, t) - V(x, t-h)}{h}$$

$h \rightarrow 0$

$$\frac{f(x, t) - f(x, t-h)}{h} = \int_0^1 \frac{\partial f}{\partial t}(x, t-h\theta) d\theta$$

where  $x^*$  is some point in the interval connecting  
 $\hookrightarrow x_h$  to  $x$ . Using (3.12)

it follows that, for any  $s > 0$ ,  $n \setminus d$

$$(r) \rightarrow \int_0^r \frac{1}{t^{s+1}} dt < \frac{1}{r^s} \quad \text{for } r > 0$$

provided  $t \rightarrow 0$  is sufficiently small (independently of  
 $\hookrightarrow x$ ). Once  $t$  is fixed  
 $=$  that if  $r < \delta$ ,  $\delta$  PS D,  
 $(ie < t)$  we can find  $d$   $d(\delta, e)$  such

$$(3.16) \quad \int_0^{\delta} \frac{1}{t^{s+1}} dt < \frac{1}{\delta^s}$$

provided  $\delta < d$ .

Writing  $\int$  in the form

and using (3.15), (3.16), we get  $\int_0^{\delta} \frac{1}{t^{s+1}} dt < 3e$  if  $\delta < \delta_0$   
 $\hookrightarrow$  8. Thus  $dV/dx$  exists  
and is equal to  $V^*$ , i.e., (3.6) holds.

**Theorem 4.** Let  $f(x, t)$  be a continuous function in  $O$  and locally Hölder

continuous with exponent  $b$  in  $i(u, t)$  uniformly with respect to  $t$ . Then  $V(x, t)$

has second continuous derivatives with respect to  $x$ , for  $x \in D$ ,  $T_0 < t < T_1$  and

$$\begin{aligned} & \left( \frac{\partial^2 V}{\partial x_i \partial x_j} \right) = \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & \rightarrow \left( \frac{\partial^2 V}{\partial x_i \partial x_j} \right) = \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau, \end{aligned}$$

$\frac{\partial^2 V}{\partial x_i \partial x_j}$

Observe that in (3.17) the order of integration is immaterial

whereas since the singularity of the integrand is absolutely integrable in each

variable separately (by (3.7)), the integral in (3.17) is a repeated integral

and thus only the  $t$ -integral is taken as an improper integral. In contrast

to (3.7) we now have the inequality (whose proof is similar to that of

$$\begin{aligned} & (3.7)), \\ & \text{th2} \quad \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & (3.18) \quad \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & \rightarrow \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & \frac{\partial^2 V}{\partial x_i \partial x_j} = \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \end{aligned}$$

so that we assert that the singularity

cannot be removed of the integrand of (3.17) is

absolutely integrable in  $O$ .

*Proof.* By Theorem 3,

where  $J$  is defined by (3.8). Write  $dJ/dx_i$  in the form

$G =$

$$\begin{aligned} & (3.20) \quad \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & \rightarrow \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \\ & \frac{dJ}{dx_i} = \int_{T_0}^t \int_D \frac{\partial^2 f}{\partial x_i \partial x_j} (x, \tau) d\tau \end{aligned}$$

|          |    |    |                |
|----------|----|----|----------------|
| + KU,    | r) | r) | <<*'(*, t;t,r) |
| fD [G({, | ^- |    |                |

$$-C(y, T) - \hat{r}_{wy}'(x, t; t, T) \sim dt$$

|   |                       |               |
|---|-----------------------|---------------|
| + | Z(x, i; {, t) [(x, r) | / (?/,r)] <ft |
|---|-----------------------|---------------|

where  $u$  is a fixed point. Let  $B_{\rho}/PS$  be a fixed ball contained in  $D$  and denote by  $\partial B_{\rho}/PS$  the boundary of  $K$  and by  $K^*$  the complement of  $K$  in  $D$ . Breaking the first integral on the right-hand side of (3.20) into two integrals, namely,

$\frac{1}{\Gamma(s)} = \frac{1}{\Gamma(s)} - J - \frac{1}{\Gamma(s)}$  and the divergence  
 $\hookrightarrow$  theorem to the first integral,

applying we obtain

$$(3.21) \quad \begin{aligned} & \text{wv}\{-x, t; e, t\} \cos \{n, \text{shch}\} \, dSv + \\ \hookrightarrow & \quad \text{wv}\{x, t; x, t\} \, ax \end{aligned}$$

$\mathbf{j} \cdot d\mathbf{K} = \mathbf{f} \cdot \mathbf{n} dS$  where  $\mathbf{n}$  is the outwardly directed normal to  $d\mathbf{K}$  and  $dS$  is the surface

element on  $dK$ . Substituting (3.21) into (3.20), then differentiating (3.20)

once with respect to  $x$ , and choosing  $u = kh$ , we get

$$W(x, I 1, t) \cos 0, 1, , :) \\ \backslash \text{JBK} \quad -^{\wedge} \quad \langle K, J \quad -^{\wedge}$$

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kh be a fixed point lying in the interior of K. Then Let each of the

first two integrals on the right-hand side of (3.22) is a bounded function of its variables which tends uniformly to zero if  $t \rightarrow \infty$ . To estimate the last integral  $I_4$  on the right-hand side of (3.22), we

↪ use (3.18) and the

Holder continuity of  $\beta$ . We find that

$\text{it } I \sim \text{const}, \quad g \quad ax \quad \text{const.}$   
 $1/4 \sim - \quad JD \quad - \quad - \quad -$   
 $(i \quad r)" \quad \backslash x \quad x|e+2-2m-0 \quad (i \quad tu$   
 if  $1 - - \quad (/0/2) < m < 1$ , where the constant is independent  
 $\hookrightarrow$  of  $(x, t, t)$

provided  $kh$  is restricted to a closed subset of  $D$ .

To evaluate the third integral  $/3$  on the right-hand  
 $\hookrightarrow$  side of (3.22) we

need the explicit formula

$X \quad - \quad - \quad PS \quad a*$   
 $\hookrightarrow (u, r)(xk -$   
 $\backslash -2\{t \quad t)a*(u, t) + \quad S \quad aih(y, r)(xh \quad \&)$   
 $\hookrightarrow \quad \&)]*$

Using (2.2), (2.3), and the mean value theorem we find  
 $\hookrightarrow$  that

$I \quad PL \quad 4(<<-r)J \quad \text{exp}L \quad 4(<<-g)L$

Using this inequality and employing once more (2.2),  
 $\hookrightarrow$  (2.3) we find from  
 the formula (3.23) that

(3.24)  $E \quad \wedge<*.':$   
 $dXi \, dx \quad b')-[i5i \quad "***<*\#'^1$   
 $< C0nSt. \{t - \quad r)-("/2)-1$

$< \quad \text{const}, (t - \quad \sim$   
 $r)^{-\wedge)-1 \quad \text{exp} \quad \backslash kh \quad *!">$   
 $[-14deg^G^|2]$

for any  $1^{\wedge} < \quad 10$ , from which it follows (as in the proof  
 $\hookrightarrow$  of (3.3), (3.7))

that

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 $\hookrightarrow \quad I'HoltLI'.M \quad (IILG. I$

$$\begin{aligned} & \text{II} < \\ (3.25) \quad & \frac{1}{(a/2)} \sum_{i=1}^n \frac{1}{(i)} \end{aligned}$$

$\sim$

Since  $|(PS, t)| < \text{const.} \cdot |x|$ , using (3.18) we conclude that the integrand of (3.25) is also bounded by the right-hand side of (3.25) (with a different constant). Hence  $|f_3| < \text{const.} \cdot (t-r)^{-1}$  if  $1/(a/2) < m < 1$ .

Since the same bound was established for  $f_4$ , and since, as has already been noted, the first two terms on the right-hand side of (3.22) are bounded functions of  $(t, t)$ , we conclude that

$$(3.26) \quad |J(x, t, t)| \leq \text{const.} \cdot L \cdot d \cdot \frac{1}{|x|}$$

$dX_i dx$

where  $d = \min_{x \in [a, b]} |f(x)|$ .

We can now proceed as in the proof of Theorem 3; namely, we set

and prove that the integral (3.19) satisfies

$sx, |ex_j|$

i.e., (3.17) holds. The continuity of  $d^2V/dx-i dx$ , follows by applying the method of proof of Lemma 1 to the right-hand side of (3.17).

**Theorem 5.** Let  $f(x, t)$  be as in Theorem 4- Then  $dV(x, t)/dt$  exists and

is continuous for  $kh \leq t \leq T_0 + I < 7\lambda$ , and

$r \gg n, t$

$$\begin{aligned} & \frac{dV(x, t)}{dt} \\ \hookrightarrow & \frac{dZ(x, t; PS, t)}{dt} \end{aligned}$$

*Proof.* Since  $Z(x, i; x, r)$  is a solution of (2.7), if  $t > t$

$$(3.28) \quad = \frac{Z(x, i; x, t)}{t, r} \frac{\partial x}{\partial t}$$

| J(x, fD I  
i,y=l J u oli d.lj  
Each on the right-hand

term side can be treated similarly  
↪ to  $d^2J/dk^2 \frac{dx}{dt}$ ,  
in the proof of Theorem 4. Hence we get (compare (3.20))

d

(3.29)  $J\{x > l > T\}$   
at  $\frac{d}{dt} J(x, t)$   
We shall prove that  $dV(x, t)/dt$  exists and

$$(3.30) \quad = J(x, t, t) + \int_t^{t+h} J(x, t, r) dr.$$

Taking  $h > 0$  we consider the finite difference

! i\* .' { |<im><\m1 m \i, <>u iidnn \m> nil. <#lin #11\  
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h

$$= \frac{J(x, t + h, r)}{h} (It + \int_t^{t+h} J(x, t^*, r) dr$$

\ R" jl ft

where  $t < t^* < t + h$ . As  $h \rightarrow 0$  the first term  
↪ on the right-hand side

converges to  $J(x, t, t)$ . To evaluate the second term, consider

Using (3.29) it follows that, for any  $\epsilon > 0$ ,

'I th

$$(3.33) \quad \frac{d}{dt} J(x, t) = J(x, t, t) + \int_t^{t+h} J(x, t^*, r) dr$$

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for some  $t_t < t$  and for all  $t < I < t - h$  provided  $h$  is sufficiently small, say  $h < d_i$ . Since  $dJ(x, t, t)/dI$  is a continuous function if  $t - r > t - t_t$ , there exists a  $d < d_i$  such that if  $L < d$  then

$$(3.34) \quad |j(MV) - |/u-, ^) \quad Tu- \quad 2U \quad <$$

for all  $t < t_t$ .

Breaking each of the integrals in (3.32) into two  $\hookrightarrow$  integrals by

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and using (3.33), (3.34), it follows that  $\|II\| < 3c$  if  $h < d$ . Using  $\hookrightarrow$  this in (3.31), we obtain (3.30) for the right-hand  $\sim$ -derivative. The

for  $h < 0$  are similar. considerations

In view of (2.10) and the fact that  $J\{x, t, t\}$  is a solution of (2.7),  $\hookrightarrow$  (3.30) reduces to (3.27). The continuity of  $dV(x, t)/dt$  follows by applying  $\hookrightarrow$  the

method of proof of Lemma I.

Combining Theorems 4, 5 we obtain:

**Theorem 6.** Let  $f(x, t)$  be os in Theorem 4- Then  $V\{x, t\}$  satisfies the

equation  $(.r \text{ PS } D, T_0 < t < I\backslash)$

No. -

$$(3.35) \quad X_{i, (, , t)} \quad \deg X_i \, dX_j \quad 2VP S D dt$$

$$\hookrightarrow \quad p > \quad \backslash$$

= -

$$-m \quad t < < < i * . o \quad \deg < < < -$$

$$*> + / e / , , \quad J, \quad \hat{a}' 1 * ; \hat{t} ) ^ t *$$

Note that the singularity of the integrand on the right-hand side of

(3.35) is absolutely and separately integrable in PS and  $r$ .

## 4 Construction of Fundamental Solutions

We shall construct a fundamental solution  
 $\hookrightarrow$   $G(gg, PS; PS, r)$  of  $Lu = 0$  in  
the form (2.8). If  $Ph$  is such that Theorem 6  
 $\hookrightarrow$  applies for  $(zh, t) = Ph(kh, t; x, g)$

then the equation  $LT = 0$  becomes

$$(4.1) \quad \begin{aligned} &= \int_{\{t\}}^{t} Ph(kh, i; x, t) LZ(x, s; x, t) dydc. \\ &\hookrightarrow \int_{\{t\}}^{t} Ph(kh, i; x, t) LZ(x, t; y, s) - F(u, f * fDLZ(x, \end{aligned}$$

Thus, for each fixed  $(x, t)$   $F(zh, i; x, t)$  is a solution  
 $\hookrightarrow$  of a Volterra integral  
equation with a singular kernel  $LZ(x, l; y, a)$ .

From  
 $= -$

$$(4.2) \quad \begin{aligned} &LZ(x, I; y, a) [au(zh, i) a << (i/, \\ &\hookrightarrow s)] Ia(kh, i; 2/, << g) \\ &2^{\sim} \quad \quad \quad \sim 7^{\sim} 7 \end{aligned}$$

$\sim \sim$

$$\begin{aligned} &+ \sum_{i=1}^6 S_{6, (r, i)} Z(kh, i; 2/, s) + c(x, t) Z(x, t; u, s) \\ &, = 1 \quad dXi \end{aligned}$$

we obtain the inequality

$$(4.3) \quad \begin{aligned} &|M(, , (*, )| <^{\sim} F - jifi - 5 - ; \\ &\hookrightarrow (i - | < m < i). \end{aligned}$$

Hence the singularity is integrable.

We shall prove that there exists a solution  $Ph$  of  
 $\hookrightarrow$  (4.1) of the form

$$(4.4) \quad \begin{aligned} &Ph(kh, t; \{, T) = S (LZ)i(z, \\ &\hookrightarrow t; x, t), \end{aligned}$$

where  $= LZ$  and  
 $(LZ)i$

$$(4.5) \quad (LZ)_{,,+i(z, \quad i; \{, t) \quad \langle r; x, t) \text{ rfy } Lg. \\ \hookrightarrow a)(LZUy, \quad [LZ(z, \quad I; y, \\ // \ /0$$

We shall prove the convergence of the series in (4.4). The following elementary lemma will be needed.

**Lemma 2.** If  $G$  is a bounded domain in  $\mathbb{R}^n$  and  $0 < a < n$ ,  $0 < b < e$ , then for any  $kh$  PS  $G$ ,  $z \in G$ ,  $\rho_0 \wedge z$ ,

$$G \quad dp \quad fconst. \quad |.r-- \quad z \setminus n \sim a \text{-deg} \\ \hookrightarrow \quad \text{if } a + \quad b > e,$$

$<-$

$$Jg \setminus kh \quad -- \quad - \\ \hookrightarrow \quad \text{if } a + \\ y \setminus a \setminus y \quad z \setminus th \quad [.const, \quad b \\ \hookrightarrow < p.$$

The proof is obtained by breaking  $G$  into three sets corresponding to

$$\setminus y \quad z \setminus < \setminus x \quad z \setminus /2, \setminus y \quad x \setminus < \setminus x \quad z \setminus /2 \text{ and the complementary set, and}$$

estimating the corresponding integrals separately.

$$\text{Using Lemma 2 and (4.3) we get, if } 2m < \quad 1 \text{ and } 2(n + \\ \hookrightarrow \quad 2 -- 2m -- a) < n,$$

$$|(LZ)^2(r, t; PS, t) \setminus < \quad - \\ \hookrightarrow \quad (2 - Im. - \langle *)' \\ \wedge \quad tu + its - T) \quad m. "IG \wedge i7 + 2 \sim 2m - ** \\ \text{Since } m < \quad 1, 2 < 2m + a, \text{ the singularity of } (LZ)^2 \text{ is weaker than that of } LZ. \\ \text{Proceeding similarly to evaluate } (LZ)^3, (/>Z)^4, \text{ etc., we arrive at some} \\ \hookrightarrow i > 0 \text{ for}$$

which

$$(*1.G >) \quad (LZ)_{,,Xx, \quad t; x, t) \setminus < const.$$

We proceed to prove by induction on  $m$  that

(4.7)  $\Gamma(m+n)(x, t; x, t) \leq$   
 $\rightarrow$   $\frac{1}{\Gamma(m+n)} \int_0^t (t-s)^{m+n-1} ds$   
 where  $K_c, K$  are some constants and  $\Gamma(i)$  is the gamma function. For  $m = 0$   
 this follows from (4.6). Assuming now that (4.7) holds for some integer  
 $m > 0$  and using (4.3) we get

$K_m$

$\Gamma(m+1+n)(x, t; x, t) \leq \text{const. } K_0$   
 $p, \quad d \backslash t \quad + \quad 1 \backslash$

$\ast V \quad - \quad -$   
 $(I \quad s) - (s \quad t) \sim as.$

$-$   $-$   
 Substituting  $r = (s \quad r)/(t \quad r)$  and using  
 $\rightarrow$  the formula

$= \ll f_c$   
 $/; (i-p) \deg v - w/P \quad T(a + \quad 6)$

(4.7) follows for  $m + .1$ , provided the constant  $j_K$  is appropriately chosen.  
 From (4.7) it follows that the series expansion of  $\Phi(t, t; x, t)$  is

convergent and that the integral in (4.1) is equal to

$S; \quad \Gamma(x, \quad t; y, c) \ast \quad (\Gamma) \chi y, \quad s; x, T) \, dy \, da.$

$J' \, J_D$

Hence  $\Phi$  is a solution of (4.1). We also have

$/, \quad -, 4 \, i w \quad \dots w^{\quad} \quad \text{const.} \quad 1$   
 $\rightarrow \quad /, \quad - \quad a \quad ^{\quad} \quad , \backslash \ast$   
 (4.8)  $|\Phi(kh, I; x, t)| \leq \quad -$   
 $\rightarrow \quad 2$

$< m <$

$-\{GZG^{\quad} \quad |g \quad \{ |i, +2.2m_a \quad ^{\quad} 1$   
 $\rightarrow 1J$

In order to study  $\Phi$  in more detail we shall need the following lemma.  
 $e - c$

**Lemma 3.** //  $-\deg \deg \ast < -1, -oc < 0 < +1$ , then

*Proof.* Substitute



const.

-

$e \in G \setminus kh \text{ SiE} <$

-  
 $(t \quad TU \setminus Kh \quad x | " \quad \exp L - i (^T J$   
 $\cdot \quad \text{const.} \quad G \quad -$   
 $d * Z(x, f, t; , r) \quad K \setminus x \quad S121$   
 (4.12)  
 $dx \ i \ dXj \ dX/,$

const.

g

-

$x_{sia} * \sin$

(4.13)  $|Sh^*, \quad i; x, r)| <$   
 $(f \quad ^ \quad ^{e+2_2m-} \quad \exp r " i(r^J$   
 for any  $XS < lo$ ; in (4.10)  $0 < m < \{e + 1\}/2$ , in (4.11)  $0 < m < (e + 2)/2$ ,  
 sh id in (4.13)  $0 < m < (w + 2 - a)/2$ .

-

Using (4.13) with  $m = (e - +- 2 \quad a)/2$  and applying Lemma 3, we find

that

const.  $G \quad --$   
 $i/Try \setminus , , i m ^ \quad \% \setminus x \quad PS | 2I$   
 $|(LZ)2(.r, t; x, g)! < \quad _ \quad ekhr$   
 $(f \quad t)(i+2-2a)/2 \quad [-4(^_t) \quad J'$

Proceeding by induction one easily establishes, with the aid of Lemma 3, the inequalities

(4.14)  $Kb^c.ijj.ol$   
 $< ^ ( , - r ) -- . . < - . \exp f - ^ f J - * ]$

where  $H_0, E$  are some positive constants. From the definition of  $Ph$  we then conclude that

From this inequality it also follows (compare  
 $\hookrightarrow$  the derivation of (4.9)) that

$i -, \quad \text{const.} \quad G$   
 $\hookrightarrow \quad --$   
 $/, , e \quad . u n, ^$   
 $\hookrightarrow \quad lo | i * \quad PS | 2 \sim I$   
 (4.10)'  $' " - <$

' V ' I, ' x,t)\

\Ph(kh,  
+ 2-2m-a exp^ ---77; -- G  
(i TU\Kh x|>>> |\_ 4(i t) J  
for 0 < m < (e + 2 -- a)/2, where Xf> is any number  
↪ less than the 15  
appearing in (4.13) and, consequently, it can be taken  
↪ to be any number  
</V (4.10) is an improvement of (4.8).

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Py using (4.1 i) and employing the; method of proof of Lemma 1, Sec. 3,  
one can show that the (LZ)v(x, t; x, t) are continuous functions of (re,t),  
with respect to (PS, t) if t -- t > const. > 0, and are continuous

uniformly

functions of (x, t), uniformly with respect to (re,t) if t -- t > const. > 0.  
It follows that the (LZ),,(rr, t; x, t) are continuous functions of (re,t; x, t).  
From (4.4), (4.14) one then concludes that F(gs, t; x, t) is also a continuous

function of (re,i; x, r).

**Theorem 7.** F(gs, t; x, g) is Holder continuous in x; more precisely, for  
any  $0 < b < a$ , -

(4.17) \Ph(kh, t; t, t) F(u, t; t, t)\

\* + -R  
0^>> {<<P No. fl  
↪ [-^}  
giLege u = a -- b and where l\* is a positive constant.

*Proof.* We first prove the inequality

(4.18) \LZ(x,t;t,T)-LZ(y,t;t,T)\  
const. \x  
↪ -  
/ps-ei , ti'-n  
-,/!>>;cxpr exp r\_%

where k is a positive constant. Consider first the case where - -

$$(4.19) \quad \sqrt{x} \quad 7/|2 < t \quad r,$$

and take the term  
 $= -$

$$(4.20) \quad F(x, t; x, r) \quad i) \quad Z\{z, t; x, r) \\ [a, -, -Cr, \quad a, (x, T)]$$

$$\widetilde{\phantom{x}}_r$$

of  $LZ\{x, \quad t-, x, t)$  (see (4,2)). We have

$$(4.21)$$

$$- - \sim$$

$$Fix, t; x, r) \quad /kh, /, f; x, t) = \quad K(.r, \quad i) \quad ats\{u, \\ \hookrightarrow t)] \quad Z(.r, i; x, r)$$

$$*_-$$

$$[a,n0/, 0 \quad b<<(x, t)]$$

$$= L + L. =$$

Using (4.11) with  $m \quad (/t + \quad 2)/2$  we get

$$(:oust* l-r" \wedge \quad - l,*|kh$$

$$\sim$$

$$(4.22) \quad Kl.11) \quad |/I| S<$$

$$\backslash F*I \text{ exD slr } G *l'1 * \_ \_$$

$$\wedge \quad t)(p+2)/2 \quad [\wedge 4(i \quad r) J$$

Using the mean value theorem and

(4.12), and noting that, because of  
 (4.ID), for any point  $z$  in the interval  $(.>>*,$   
 $//)$

```

cxijG~c                                     tZ
↪      1      (foranydeg<c'<c>)>
t.Z*'"]^('degnt-exp\~c'

```

we m;ili  
92 - ~ ~

$$(4.23) \quad Z(x, t; x, g) \quad Z(y, t; x, r)$$

$$dX_i \quad dx_j \quad du_g \quad d\pi_{ij}$$

$$\text{const. } \backslash x - y \backslash \quad G \quad -$$

$$\backslash y \quad \{ [P$$

S - P

(i)  $r(-+3)/2$  L t-rj

where  $\&\gg$  are used to denote appropriate positive constants. Hence,

$$(4-24) \quad |f_t| < \exp \left[ \sum_{j=1}^n k^j f_j \right]$$

Combining (4.24), (4.22) and using (4.19), we get

$$\frac{\lambda F(x, t; x, r)}{(tm)T^{S}} \quad F(a, t; i, T) | < \quad \exp \quad [-*, \quad l^{f*}]$$

The estimation of the Holder exponent and coefficient for the lower-order terms in LZ is similar to the estimation for F. Combining these estimates,

the inequality (4.18) follows.  
 If (4.19) is not satisfied then, by (4.13) (with  
 $\hookrightarrow \quad m = (e + 2 - \langle x \rangle / 2),$

$\sim \sim$

$$\begin{aligned} & \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} f(s) ds \\ &= \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} \left[ \sum_{k=0}^\infty \frac{(-1)^k}{k!} \frac{\partial^k f}{\partial x^k}(s) \right] ds \\ &= \sum_{k=0}^\infty \frac{(-1)^k}{k!} \frac{\partial^k f}{\partial x^k}(t) \int_0^t (t-s)^{-\alpha} ds \\ &= \sum_{k=0}^\infty \frac{(-1)^k}{k!} \frac{\partial^k f}{\partial x^k}(t) \frac{t^{1-\alpha}}{1-\alpha} \end{aligned}$$

--  
 A similar inequality holds for  $LZ(y, t; x, t)$ . Since  $(Z - t)b/'z < \backslash x - t/I$ ,  
 (4.18) follows.  
 Denoting the integral on the right-hand side of (4.1) by  $\hat{(\cdot, t; x, t)}$ ,  
 it remains to prove that (4.18) holds with  $LZ$  replaced  
 $\hookrightarrow$  by  $Ps$  (with a possibly

different  $k$ ). Writing  
 --

$Y = (z, i; x, t) \quad '(u, t; x, t) - \quad t; f, s)$   
 $\hookrightarrow LZ(y, \quad t; f, s)]$   
 $f * fD [LZ(x,$

$* Ph(z, s; x, t), df da$

and using (4.18), (4.15), we get

(4.25)  $|*(M; x, t) - Ps_0 / , i; x, t)|$   
 $< \text{const.} \backslash x - \quad u \backslash "[1(kh, t; x, t) 4 - / (//, i; x, t)]$

where

$/(i, <; x, t) \quad + 2 - U) / 2 \quad ThKhR [$   
 $\hookrightarrow \quad + 2 - c - <<) / 2$   
 $JT )b (i \_ \quad s)(e \quad - \quad f \_ s \quad J (s$   
 $\hookrightarrow \quad r)(n$

By Lemma 3,

$J(*> <<*L, t)< \quad - \quad \exp \quad - / . * <$   
 $(t \quad TyH \sim z^i \quad [ \quad UKs - f \quad ]'$

A similar inequality holds for  $I(y, t \backslash PS, t)$ . {Substituting  
 $\hookrightarrow$  these inequalities

into (1.2G) we  $IiikI I, Ii:i.I,$   
 $(I.IK)$  is satisfied if  $/, X$  is replaced by  $ps$  and if  $k$

is  $K'$ phu-ed by A-4. This completes the proof of Theorem 7.

It's reading carefully the proof of Theorem 7 one deduces:

**Corollary.** The inequality (4.17) holds for any  $l^* < lo/4$ .

**Theorem 8.** The function  $G(zh, t; x, r)$ , defined by (2.8), is a fundamental  
 aviation of  $Lu - 0$  in  $O$ .

*Proof.* We first prove that for each fixed  $(PS, t)$   $LT = 0$ . Write  $G$  in the form

$$(4.2C) \quad \frac{\partial}{\partial t} Z(x, t; x, t) + \frac{\partial}{\partial x} Z(x, t; v, s) \text{Ph}(i7, <r; x, r) \text{ae da}$$

$T(x, t; x, t) \text{ fD f}^*$

$$+ \frac{\partial}{\partial t} Z(xt t; e, s) \text{Ph}(e, a; x, t) \text{ae da},$$

$\hat{J}_0$

where  $t_0$  is a fixed number satisfying  $t < t_0 < t$ . For the first integral, the

first two  $z$ -derivatives of  $Z(x, t \setminus t \setminus, a)$  are continuous functions in  $(z, t; e, a)$  whereas  $\text{Ph}(e, a; x, t)$  is absolutely integrable in  $(17, s)$  (by (4.8) or (4.16)).

Hence, by a standard theorem of calculus, the first two  $x$ -derivatives of the

integral exist, and the order of any  $\hat{\cdot}$ -differentiation (up to the second order)

and of the integration may be changed.

As for the second integral on the right-hand side of (4.26),  $\text{Ph}(e, a; x, t)$  (for  $(x, t)$  fixed) is uniformly Holder continuous in  $e$  with any exponent  $< a$ , as follows by Theorem 7. By Theorems 3, 4 of Sec. 3 it follows that the first two  $t$ -derivatives of the integral exist, and the order- of any  $t$ -differentiation (up to the second order) and of the integration may be changed. We conclude that the first two  $z$ -derivatives of  $G$  exist, and

a similar formula holds for  $\text{th}G/\text{th}^{**},$ . The existence of  $dT(x, t; x, t)/\text{th}i$  can be established by the same

considerations as for  $dT/dKhg, d2G/dkh, dkh3$ , making use of Theorem 5, Sec. 3. The formula

$$(4.28) \quad \frac{\partial}{\partial t} T = \frac{\partial}{\partial x} Z(x, t; v, s) \text{Ph}(e, a; x, t) \text{ae da} + \frac{\partial}{\partial t} Z(x, t; v, s) \text{Ph}(e, a; x, t) \text{ae da}$$

$\sim \text{''} C)$

$$*(, < < ! 6 t) + \frac{\partial}{\partial t} Z(x, t; v, s) \text{Ph}(e, a; x, t) \text{ae da}$$

is valid.

Combining (4.28), (4.27) and the analogue of (4.27)  
 $\hookrightarrow$  for  $dT/dx_h$  we get

"L) IAI. ;(>1,IPIONS LM>  
 $I \cdot I)N| >> LMI - ' . N$  I'

111 CUKh GIMMILILI GIIL'I

-

$LV(x, t; x, r) = LZ\{x, I; x, t\} F(1; I; x, t)$

+

/t /D LZ^'  
 $\wedge 4 >> s) \cdot ph^'$  s5 & t) d-e da.

Since  $\Phi_h$  satisfies the integral equation (4.1),  $LT = 0$ .

Using (2.8) and the method of proof of Lemma 1, Sec. 3, one can show

that  $G(.t, t; x, r)$  is a continuous function of  $(x, t)$ , uniform] u with respect  
to  $(x, t)$  if  $t - r > \text{const.} > 0$ , and it is a continuous function of  
 $\hookrightarrow (PS, r)$   
with to if  $t - r > \text{const.}$  Hence  
uniformly respect  $\{x, t\}$   $G(kh, i; x, r)$   
 $\hookrightarrow$  is a  
continuous function of  $(kh, PS; x, r)$ . Here  $kh$  and  $x$  vary in  $D$  and  $T_0 < t <$

$i < I \setminus$ .

(4.28), (4.27) and  
Using the analogue of (4.27)  
 $\hookrightarrow$  for  $dT/dx_i$  one can

similarly show that  
 $thG(.t, t; g, r)$

$dT(T, t; t, T)^\wedge aT(y, \wedge, r);$

are continuous functions of  $(x, i; x, t)$ , where  $kh, x$  vary in  $/)$  and  $7T_0 < t <$

$t < T_u$ . This fact, however, will not be needed in the present chapter.

It remains to prove that for any continuous function  $/(.r)$  in  $D$ ,

$$(4.29) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

In view of Theorem 1, Sec. 2, it suffices to show that

$$(4.30) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

for  $J^* \in \mathbb{R}^n$

$$(4.31) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

$$(4.32) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

$$(4.33) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

## 5 Properties of Fundamental Solutions

In Sec. 3 we have the concept introduced of volume potentials with respect to the parametrix  $Z(x, t; x, t)$  and established some differentiability

$$(5.1) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

a similar study of volume potentials with respect to the fundamental  $G(h, I; x, t)$ . Thus, we shall consider functions of the form solution

$$(5.2) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

for  $J^* \in \mathbb{R}^n$

$$(5.3) \quad \lim_{\substack{h \rightarrow 0 \\ \text{for } h \in \mathbb{R}^n}} \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} = \frac{G(h, t; \mathcal{P}, r)}{(\mathcal{P})} \quad \text{for } h \in \mathbb{R}^n$$

for  $J^* \in \mathbb{R}^n$

If we substitute for  $G$  its expression from (2.8), then we find that

$$(5.2) \quad W(x, t) = V(x, t) + \int_0^t \int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |x| \leq r\}} \Delta u(x, s) dx ds$$

where  $V(x, t)$  is the potential (3.1) and  $U(x, t)$  can be written (after

the order of integration) in the form changing

$$=$$

$$(5.3) \quad U(r, t) = \int_0^t \int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |x| \leq r\}} \Delta u(x, s) dx ds$$

in  $J_D$  where

$$(5.4) \quad = \int_0^t \int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |x| \leq r\}} \Delta u(x, s) dx ds$$

$f(t, x)$

$$= 0 \text{ if } f < t \text{ and } (4.8), \text{ it follows}$$

that

Defining  $\Phi(k, i; x, g)$  recalling

**Lemma 1, Sec. 3**, can be applied to the right-hand side of (5.4). Hence  $f_a(t)$  is a continuous function.  $f$  is also uniformly Hölder continuous in  $k$

with any exponent  $b < a$ . Indeed, using (4.17) we get

--

$$(5.5) \quad \int_0^t \int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |x| \leq r\}} \Delta u(x, s) dx ds \leq \text{const.} \cdot |x|$$

that

where

$$A(x, t) = \int_0^t \int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |x| \leq r\}} \Delta u(x, s) dx ds$$

Substituting (for  $t$  fixed)  $r = |x|$  we find that

$$A(x, t) \leq \text{const.} \cdot G(t - g(t-2)/2) \leq \text{const.}$$

To  $A$  a similar inequality holds for  $A(y, t)$ . Substituting these inequalities into

(5.5), the uniform Hölder continuity (exponent  $\alpha$ ) of  $f(x, t)$  in  $x$  follows.

**Theorem 9.** //  $f(x, t)$  is a continuous function in  $O$  then  $W(x, t)$  is a

continuous function in  $O$  and  $dW/dx_i$  are continuous functions for  $x \in D$ ,  $T_0 < t < T$ . If  $f(x, t)$  is also locally Hölder continuous in  $x \in D$ , uniformly with respect to  $t$ , then  $d^2W/dx_i dx_j$  and  $dW/dt$  are continuous functions for

$x \in D$ ,  $T_0 < t < T$ , and

$$(5.6) \quad L \nabla V(x, t) = -f(x, t).$$

*Proof.* All the assertions of the theorem, except for (5.6), follow

immediately from the formulas (5.2)-(5.4) and from the Hölder continuity of  $f(x, t)$  in  $x$ , by applying Theorems 2-5 of Sec. 3. (5.6) is a consequence of Theorem 6, Sec. 3 and (4.1). Indeed,

$$LW(x, t) = -f(x, t) - f(x, t) + LZ(x, t; x, t)/(x, t) \frac{dx}{dr}$$

$$f(x, t) + J(t)$$

$$L \nabla A(x, t; e, s) \cdot (e, s) \, dV \, da$$

$$J(t) = 0 + -f(x, t)$$

$$g) [-F(x, t; x, r) + LZ(x, t; x, r)$$

$$f(x, t) \, dV \, da,$$

$$+ LZ(x, t; e, s) \cdot \Phi(e, s; x, t) \, da \quad c/PS$$

$$]t] \, dV \, da = t). -f(x, t)$$

## 6 Fundamental Solutions in Unbounded Domains

In this section we shall extend the results of Sees. 1-5 to the case where  $D$  is an unbounded domain. The special case  $D = \mathbb{R}^n$  is of particular importance. Since most of the arguments are similar to those for the case where  $D$  is bounded, we shall only describe the necessary modifications. If  $D$  is unbounded we always assume in this chapter that  $L$  satisfies

the following assumptions (which coincide with  $(A_i)$ ,  
 $\hookrightarrow$   $(A_2)$  of Sec. 1 if  $D$  is

bounded):

$(A_i)$   $L$  is uniformly parabolic in  $0 \leq t \leq T$ ;

$(A_2)$  the coefficients of  $L$  are bounded continuous functions in  $O$  and

$(1.4)$ ,  $(1.5)$ ,  $(1.6)$  hold throughout  $O$ .

The definition of a fundamental  $T(x, t; PS, t)$  is the same solution as in

(6.1)  $|f(x)| < \text{const} \cdot \exp[h|x|^2]$

for some –

positive constant  $h$ . The integral in (1.7) exists only if  $4L(PS$   
 $\hookrightarrow r) <$   
 $l_0$  (where  $l_0$  is as in (2.2)). If

(6.2)  $h <$

$shch^{\wedge}$

then the integral in (1.7) exists for all  $T_0 < r < t < T_b$   
 The inequalities (4.9)-(4.13) obviously remain true also in case  $D$   
 $\hookrightarrow$  is

unbounded. The

inequalities (4.14)-(4.16) and (4.17) also remain true

without any change in the proofs. The proof of Theorem 8, Sec. 4, can also be extended with obvious modifications, provided Theorems 2-6 of Sees. 2, 3 are valid for the case where  $D$  is unbounded. In order to prove these theorems in this case, we break integrals  $\int g(x, t; PS, r) d\%$  into two parts by dividing the domain of integration into two domains. The first domain  $D_0$  is bounded and contains  $x$ , and the complementary  $D_i$  domain is unbounded. The first integral can be treated as in Sees. 2, 3, whereas the second integral contains no singularity and can be treated by standard theorems of calculus. Thus in the proof of Theorem 1, Sec. 2, we write  $J = J' + J''$  where in  $J'$  the  $\wedge$ -integration is taken over a bounded domain  $D_0$  containing  $kh -$

such that  $|x - x_0| > 1$  if  $x \in D_0$ .  $J'$  can be treated exactly  
 $\hookrightarrow$  as  $J$  in the proof

of Theorem 1. As for  $J''$ , assuming that

$$(6.3) \quad |f(x, t)| < \text{const} \cdot \exp[u|z|^2]$$

where  $h$  satisfies (6.2) and noting that the integral

is convergent for all  $t' < t < T$  if  $l$  is sufficiently close  
 $\hookrightarrow$  to 0, and

III, furthermore,

$$\lim_{t \rightarrow 0} \int_{D_0} \frac{\partial V}{\partial x} dx = 0 \quad \text{if } r < \epsilon,$$

we conclude that  $V$  exists for all  $T_0 < r < t < T$  and is continuous,  
 $\hookrightarrow$  and  
 $J'' \rightarrow 0$  as  $t \rightarrow 0$ . The assertion of Theorem 1 then follows.  
 We proceed to extend the results of Sec. 3. In view of (6.3), (6.2),  
 (4.9), the integral (3.1) exists. The continuity of  $V(x, t)$  follows by breaking

the  $Z$ -integral into two parts and treating each part separately. The of the integral corresponding to the unbounded part  $D_1$  of  $D$  ( $x$  is bounded away from  $D_2$ ) follows by a standard theorem of calculus, whereas the continuity of the integral corresponding to the bounded part  $D_0$  of  $D$  follows by employing Lemma 1, Sec. 3. To extend Theorem 3, Sec. 3, we again break integrals  $V$  into two  $V_1 - V_2$ .  $V_1$  corresponds to  $D_0$  and is treated as in Sec. 3. As for  $V_2$ ,

the integrand has no singularity and therefore the proof that  $dV/dx$  exists and is equal to the integral of the  $\partial V / \partial x$ -derivative of the integrand

follows by a standard theorem of calculus. The continuity of  $dV/dx$  follows

from the continuity of  $dV/dx$ ,  $d^2V/dx^2$

By the same considerations we extend Theorems 4, 5 of Sec. 3. Having extended the results of Secs. 2-4 to the case where  $D$  is unbounded, the results of Sec. 3 now also extend to this case without any change in the proofs. For the sake of later reference we sum up some of the results (in the case of an arbitrary domain  $D$ ) in the following theorem.

**Theorem 10.** Let  $D$

be any domain in  $R^n$  and assume that  $(A_i)', (A_g)'$  hold. Then there exists a fundamental solution  $T(x, t; x_0, t_0)$  of  $L u = 0$  given by (2.8), (4.1). If  $f(x, t)$  is any continuous function in  $\bar{D}$  satisfying (6.3), where  $h$  satisfies (6.2), then the function  $W(x, t)$  defined in (5.1) is a uniformly

continuous function in  $O$ . If  $f(x, t)$  is also locally Holder continuous in  $x$  on  $D$ ,

$$\begin{aligned} &\text{uniformly with respect to } t, \text{ then } \frac{dW}{dx_i} \frac{d^2W}{dx_i^2} \\ \hookrightarrow \frac{dW}{dt} &\text{ exist and are} \\ \text{continuous functions for } x \in D, T_0 < t < T_h \text{ and } &LW(x, \\ \hookrightarrow t) = &-f(x, t). \\ \text{Let } (A_1)', (A_2)' \text{ hold with } D = R_n, \text{ and consider the function} \end{aligned}$$

$$(6.4) \qquad W(x, t) = \int_{\mathbb{R}^n} r(x, t; x, T_0) \varphi(x) \, dx$$

for  $x \in \mathbb{R}^n$   
where  $\varphi(x)$  is a continuous function in  $\mathbb{R}^n$  satisfying  
 $\ast$

$$(6.5) \qquad \|\varphi(x)\| < \text{const}, \exp[-\|x\|^2] \qquad \|x\| < T_0^{-1}$$

$4/T_0$

$$\begin{aligned} &\text{We can decompose } W \text{ analogously to (5.1): } W = V + \\ \hookrightarrow &U \text{ where} \end{aligned}$$

$$(6.6) \qquad V(x, t) = \int_{\mathbb{R}^n} Z(x, t; x, T_0) \varphi(x) \, dx$$

for  $x \in \mathbb{R}^n$

$$(6.7) \qquad \text{Si } t = \tau \text{, } \varphi(x) = \int_{\mathbb{R}^n} \varphi(x) \, dx$$

for  $x \in \mathbb{R}^n$ , where  
 $=$

$$(6.8) \qquad \varphi(x, t) = \int_{\mathbb{R}^n} \varphi(x, t; x, T_0) \varphi(x) \, dx$$

for  $x \in \mathbb{R}^n$

$$\begin{aligned} &\text{Using (6.5) and (4.17) (which, by the corollary to Theorem 7, holds for} \\ &\text{any } 1 \leq l \leq 4), \text{ we find that } f(x, t) \text{ is locally Holder continuous in } x, \\ &\text{uniformly with respect to } t \text{ in intervals } T_0 + s < t < T_h, s > 0. \text{ It follows that} \end{aligned}$$

$$(6.9) \qquad LW = \int_{\mathbb{R}^n} LZ(x, t; \varphi, T_0) \, dx$$

$\int_{\mathbb{R}^n} dx +$

$$\int_{T_0}^t \int_{\mathbb{R}^n} L_Z(x, t; y, s) f(u, s) dy \quad \varphi(kh, t).$$

$\mathbb{R}^n$

Substituting  $\varphi$  from (6.8) and changing the order of integration, we get

$$\begin{aligned} L_W &= \int_{T_0}^t \int_{\mathbb{R}^n} L_Z(x, t; y, s) da \quad L_Z(x, t; y, s) \\ &\hookrightarrow -F(u, s; x, G_0) dy \\ &= \int_{\mathbb{R}^n} [L_Z(x, t; y, s) - L_Z(x, t; y, s_0)] dy \end{aligned}$$

$-\varphi(kh, i; x,$

$$T_0) \} \sim$$

$= 0.$

We thus have:

**Theorem 11.** Let the assumptions  $(A_kh)'$ ,  $(A_2)'$  be satisfied with  $D = \mathbb{R}^n$ ,

and let  $\varphi(kh)$  be a continuous function in  $\mathbb{R}^n$  satisfying (6.0). Then the function  $1G$ , defined by (6.4), exists for  $T_0 < t < T_1$  and satisfies:

$$(6.10) \quad \int_{\mathbb{R}^n} L \setminus V\{x, t; y, s\} dy = 0 \quad \text{for } T_0 < t < T_1$$

$$(6.11) \quad \int_{\mathbb{R}^n} L \setminus V\{x, t; y, s\} dy \rightarrow \varphi(kh) \quad \text{as } t \rightarrow T_0.$$

The following inequalities will be needed later on:

$$(6.12) \quad \int_{\mathbb{R}^n} L \setminus V\{x, t; PS, r\} dy < \text{const}, (1 - r)^{-1} \exp$$

$dT(x, t; PS, t)$

$$(6.13) \quad \int_{\mathbb{R}^n} L \setminus V\{x, t; PS, r\} dy < \text{const}, (1 - r)^{-(n+1)/2} \exp$$

for any  $1? < 10$ .

To prove (6.12) we use (4.9) (with  $m = e/2$ ) and (4.15), and thus

obtain, with the aid of Lemma 3, See. 4, a bound on the integral on the

right-hand side of (2.8). Combining this bound with (4.9) (for  $m = te/2$ ), and recalling (2.8), the inequality (6.12) follows. (6.13) follows in a similar manner, making use also of (4.10) (with  $m = (e + 1)/2$ ).

We conclude this section with a remark concerning the condition (1.4). In Chap. 9 we shall construct a fundamental solution for parabolic systems of any order. If we specialize the results to the parabolic equation (1.2)

then we find that all the results of Sees. 2-6 are valid also if (1.4) is

replaced by the weaker condition  
 SK.C.7  $\min_{t \in [0, T]} \inf_{x \in \mathbb{R}^n} \lambda(x, t) > 0$

K  $\lambda(x, t) \geq \lambda_0 > 0$  I'll t0 IILKM 2G>

--

(G>>.1-1)  $\lambda(x, t) \geq \lambda_0 > 0$  aph0, 01 <  
 $\hookrightarrow M^*$  Aa\*

The reason that we are able to obtain the same results under this weaker condition lies in the fact that in Chap. 9 we take for the parametrix a fundamental solution  $Z$  of parabolic equations with coefficients depending on  $t$ .  $Z$  will not be given explicitly; this will necessitate some additional analysis not encountered in the present chapter.

## 7 The Cauchy Problem

Given a function  $f(x, 0) = b(x)$  on  $[0, T]$  and a function  $\varphi(x)$  in  $\mathbb{R}^n$ ,

the problem of finding a function  $u(x, t)$  satisfying the parabolic equation

(7.1) 
$$\begin{cases} u_t - Lu = 0 & \text{in } \mathbb{R}^n \times (0, T], \\ u(x, 0) = \varphi(x) & \text{on } \mathbb{R}^n, \end{cases}$$

where  $L$  is defined in (1.2), and the initial condition

(7.2) 
$$u(x, 0) = \varphi(x) \text{ on } \mathbb{R}^n$$
  
 is called a Cauchy problem (in the strip  $0 < t < T$ ). The solution is always

required to be continuous in  $\bar{D}$ .

$f(x, t)$  and  $\varphi(x)$  will be assumed to satisfy the  
 $\hookrightarrow$  boundedness conditions

$$(7.3) \quad |f(x, 0)| < \text{const} \cdot \exp[h|x|^2],$$

$$(7.4) \quad |\varphi(kh)| < \text{const} \cdot \exp[h|x|*]$$

where  $h$  is any positive constant satisfying  
 $h < *$

$$(7-5)$$

^

**Theorem 12.** Suppose that  $L$  satisfies  $(A_k h)'$ ,  $(A_2)'$  (with  $D = \mathbb{R}^n$ ,  $T_0 = 0$ ,  
 $=$

$T \setminus T$ ) and let  $f(x, t), \varphi(x)$  be continuous functions in  $0$  and  $\mathbb{R}^n$   
 $\hookrightarrow$  respectively,  
satisfying (7.3), (7.4). Assume also that  $f(x, t)$  is locally Holder  
 $\hookrightarrow$  continuous

(exponent  $\alpha$ ) in  $k h \leq \mathbb{R}^n$ , uniformly with respect to  $t$ . Then the function  
 $= -$

$$(7.6) \quad u(x, t) = \int_{\mathbb{R}^n} G(x, t; y, 0) f(y, 0) dy \\ \hookrightarrow T(x, t; x, t) / (\varphi(x, t) \frac{d}{dt} \varphi(x, t))$$

$$j \mathbb{R}^n / J \mathbb{R}^n$$

is a solution of the Cauchy problem (7.1), (7.2) and

$$(7.7) \quad |u(x, 0)| < \text{const} \cdot [k|x|*] \\ \hookrightarrow \text{for } (x, t) \in K_s 0,$$

where  $k$  is a constant depending only on  $h, l_0, T$ .

*Proof.* In view of Theorems 10, 11 of Sec. 6, it only remains to establish

(7.7). We shall need the following lemma.

**Lemma 4.** For any  $h > 0, s > 0$  there exists a  $S = C(h, c)$  such that  
the inequality

$$(7.8) \quad |u(x, t)| < C |z|* \exp(h + e) \exp$$

holds for all  $kh \leq R_n$ ,  $PS \leq R_n$

*Proof.* If  $4 - 0.1mm \leq (7.5) \ln K_1$  for  $i \leq S > h$ . If  $PS \leq 0$   $svX \leq jr/|^\wedge|$ ,

$e = x/|x|$ . Then (7.8) takes (,ho form

--

(7.9)  $h|y| e|2 (A + c) <$   
 $\hookrightarrow C|y| \backslash$   
 Now if  $|y| \leq e|2 < (A + t)/h$  then the left-hand side is  $< 0$  and  
 $\hookrightarrow$  hence (7.9)  
 holds. If on the other hand  $|y| \leq c|2 > (h + t)/h$  then  
 $-1 = u > 0$   
 $|y| > [(h + e)/hy/2$

and (7.9) clearly holds if  $S$  is sufficiently large, depending only on  $h, u$ .

Returning to the proof of (7.7), we first obtain from  
 $\hookrightarrow$  (6.12) the inequality

$|r(*.;fc r)l \wedge$   
 $\hookrightarrow [_(L + * "*)2]$   
 $(f0 \sim \exp [- \sim if]ekhr$   
 for some  $e > 0$  sufficiently small, making use of the assumption  
 $\hookrightarrow$  (7.5).

Using this inequality and (7.3) to evaluate the second integral on the

side of (7.6) we get, after substituting  $kh \leq$

right-hand  $PS$  for  $f$ , the bound -

$\exp [-e|PS|2/(< t)] , , , 1$   
 $/ / \leq$   
 $const, + < < /' / \# / , \backslash / 9 < ^PS * " r"sup$

--

$\{ \exp [A|zh x|2 (A + 0le12)] \} < const, \exp [C|x^2],$

where use has been made of (7.8). The first integral on the right-hand side of (7.6) is estimated in the same way.

## 8 The Adjoint Equation

The adjoint equation of (1.2) is, by definition, the equation

$$(8.1) \quad \begin{aligned} & \text{LH} = && \langle \langle \cdot, \cdot \rangle, I \rangle + \\ \hookrightarrow & \text{Shchkh}, \quad t) + && \langle \langle * \langle * \cdot \rangle \rangle + && = 0 \end{aligned}$$

$J. \wedge S. g \mid$  where

$$bt = -b, - + 2 S \quad ps1,$$

(8.2)

$J \text{ flbi}, PS \text{ b2a}, i$

$$c^* = s - S \text{ t-1} + S$$

=

$$i \text{ i dXi} =$$

$ij \text{ i dXi dXj}$  Thus, we may also write  
 $L^*v = \sim \sim$

$$(8.3) \quad \begin{aligned} & S \quad \sim^Z \quad \langle W \rangle \quad S \text{ PS} \sim \\ \hookrightarrow & (M \quad +co \quad + \\ ij = & \backslash dXi dXj \quad i^{\backslash oXi} \quad at \end{aligned}$$

When we speak of  $L^*$  we always assume that  $\text{dats/dkh-n}, d2a^{\wedge}/dkhi.dkhk, dhi/dxh$

exist and are, say, continuous functions. If  $i$  and  $n$  are smooth functions in  $O$  (in this context "smoothness")

means that the first two  $\sim$ -derivatives and the  
 $\hookrightarrow$  first  $\sim$ -derivative exist and

are continuous) then, as can easily be verified,  
 $II. S \text{ I'UNIHMI MLI}, \text{„OI}, II$

$$PON-. LNI> TIK \quad ('LP1ChP \quad II\{(IKI.KM$$

$$\begin{aligned} & (\langle \langle \# \\ & \langle \langle \# \text{ i '****}, \\ Lj- \text{ i } \backslash & \quad 0xj \quad ox, \quad dXj / \quad J \end{aligned}$$

- a / ch

$J_t\{m\} -$

This identity is called Green's identity for the operator  $L$ . If  $u$  and  $v$  vanish in some neighborhood of the boundary of  $O$  then, by integrating both sides of (8.4), we get

$$(8.5) \quad \int_O \left( \frac{dx}{dt} - (vLu - uL^*v) \right) = 0.$$

fa

**Definition.** A linear partial differential operator (in the independent variables  $x_1, \dots, x_m$ ) is a finite sum

The coefficients  $a_{\alpha}$  are called the coefficients of the operator.

**Theorem 13.** //  $L^*$  exists and if  $L$  is a linear partial differential operator in the variables  $(x, t)$  with, say, continuous coefficients satisfying

$$- = 0$$

$$(8.6) \quad \int_O (vLu - uL^*v) \, dx \, dt = 0$$

for all smooth  $u, v$  which vanish in some neighborhood of the boundary of  $O$ ,

then  $L = L^*$  i.e., the coefficients of  $L$  coincide with the coefficients of  $L^*$ .

Indeed, it follows from (8.5), (8.6) that

$$- = 0$$

$$\int_O u(L^* - L)v \, dx \, dt = 0$$

for all  $u, v$  as in (8.6). Hence  $(L^* - L)v = 0$ . We can now employ the

following theorem, the proof of which is left to the reader (see Problem 2).

**Theorem 14.** //  $M$  is a linear partial differential operator in the variables

$x_1, \dots, x_m$ , with coefficients defined in a domain  $G$  and if  $Mv = 0$  for all

infinitely differentiable functions in  $G$  with a compact support, then  $M = 0$ , i.e., the coefficients of  $M$  are identically zero.

From Theorem 13 it follows that the adjoint  $L^*$  of  $L$  can be defined,

equivalently, by the condition (8.5). Writing (8.6) in the form

$$\begin{aligned} & \frac{dx}{dt} = 0 \\ & (uL^*v - vLu) \end{aligned}$$

we conclude that

$$\begin{aligned} (8.7) \quad & L^*v = 0 \\ & (L^*)^*v = Lv. \end{aligned}$$

**Definition.** A fundamental solution of  $L^*v = 0$  is a function  $G^*(x, t; \xi, \tau)$

defined for all  $(x, t) \in D$ ,  $(\xi, \tau) \in D$ ,  $\tau < t$  which  
 $\rightarrow$  satisfies the following

conditions:

- (i) for each fixed  $(\xi, \tau)$ , it satisfies the equation  
 $\rightarrow L^*v = 0$  (as a function of  $(x, t)$ ,  $(\xi, \tau) \in D$ ,  $\tau < t < r$ );
- (ii) for every continuous function  $f(x)$  in  $D$   
 $\rightarrow$  (satisfying (6.1) if  $D$  is unbounded)

$$(8.8) \quad \lim_{\tau \rightarrow t} \int_D f(x) G^*(x, t; \xi, \tau) dx = f(\xi).$$

Let  $D \subset \mathbb{R}^n$  in addition shall now specialize to the case  $D$  and assume, to  $(A_i)'$  of Sec. 6, the following condition:

$(A_3)'$  The functions

$$\begin{aligned} (8-9) \quad & a_k \\ & \rightarrow c_k \\ & e_k = f_k(x) \end{aligned}$$

are bounded continuous functions in  $O = \mathbb{R}^n \times [T_0, T_1]$ ; they satisfy a uniform Hölder condition (exponent  $\alpha$ ) in  $(x, t)$ , uniformly with respect to  $t$ , and (1.4) holds throughout  $O$ .

We can then construct a fundamental solution  
 $\rightarrow G^*$  of  $L^*v = 0$  by the

method used to construct  $G$ . Thus, for a parametrix we take  
 $= -$

$$(8.10) \quad Z^*(x, t; f, r) = C(\{, r)(r^{0-ia/2} \exp \left( \int_{-\infty}^x (-f^{\sim}) \right) \\ \hookrightarrow \text{Z* satisfies the equation } LPSu = 0 \text{ where } L0 \text{ is} \\ \hookrightarrow \text{defined in (2.7). Instead of (2.8) we now take}$$

$= +$

$$(8.11) \quad T^*(kh, i, x, t) = Z^*(x, t; t, r) \quad f_j \\ \hookrightarrow \int_{-\infty}^x f DZ^*(z, t; \dots, c) 1 > *b, c; t, T) d, da$$

where  $F^*$  is the solution of the integral equation  $= t; x, t)$

$$(8.12) \quad F^*(kh, t; x, r) = L^*Z^*(x,$$

$$+ \int_{-\infty}^x L^*Z^*(x, t; e, s) *Ph^*(e, s; f, t) ae da.$$

$f_j f_D$

An analogue of Theorem 10, Sec. 6, is valid and we also have,

$$\text{to (0.12), (6.13),}$$

analogously -

$$(8.13) \quad |G^*(kh, t; x, r)| < \text{const}, (r - t)^{-it} \exp \left( \int_{-\infty}^x (-f^{\sim}) \right) \\ \hookrightarrow [-^{\sim} *]' ]$$

$$(8.14) \quad G^*(kh, i, i, t) < \text{const}, (r - f)^{-(i+1)/2} \\ \hookrightarrow \exp \left( \int_{-\infty}^x (-f^{\sim}) \right) \\ \hookrightarrow G^{-14} [^{\sim} jf ]$$

**Theorem 15-** Under the assumptions  $(A_i)'$ ,  $(A_z)'$  a fundamental solution.

$$G^* \text{ of } L^*v = 0 \text{ exists and}$$

$$(8.15) \quad G^*(x, t; zh, i) = G(zh, f; x, t) \quad (i > g).$$

*Proof.* The existence of  $G^*$  has already been discussed. It remains to

establish (8.15). Consider the functions

$$g(x, s; PS, t),$$

for  $x \in L$ ,  $t < s < t$ . Integrating (Irc-on's identity (8.4) over the domain  $-e \in (\# > 0, t > 0)$  and using  $Lu = 0$ ,  $L^*i^* = 0$  we

obtain

$$(8.16) \quad f \in R \cup \{y, t\}$$

$J \setminus m$

$$e) \nabla \{y, t\} - e) dy$$

$\sim$

$$U \in JJ, t + e) \nabla(y, T + e) dI = I t, R$$

$j \in R$  where

$$\begin{aligned} & r \quad i. / \text{'-} \ll / * \quad G \wedge / \quad \text{th} M \quad Eu \text{ --} \quad \text{ths, -} D, \\ & \hookrightarrow, \quad I \\ & Km = S / , / , , , \quad 2, ( \text{'-} \ll ij \wedge \text{ --} \quad \text{shi, --} \quad \text{sh-s-} \wedge l \\ & \hookrightarrow + o, ? \ll * \\ & , * = i \text{ Ug+}, U! g / ! = l > [ \_ J =, \backslash \quad \text{ay, -} \quad J \%, - \quad \text{ay, -} / \quad J \end{aligned}$$

$* \cos$

$(i, y_i) dS$ , do,  
 $n$  is the outwardly directed normal to  $\partial \Omega = R$ , and  $dS_i$  is the surface element on  $\partial \Omega = R$ . Using (0.12), (C.13) and (8.13), (8.14) we find that  $f t, n \rightarrow 0$  as  $R \rightarrow \infty$ . Hence,

Taking  $e \rightarrow U$  and using the second property in the definition  
 $\hookrightarrow$  of

solutions we get  $u(x, t)$

$$\text{fundamental} \quad = n(x, t), \text{ i.e., (8.15) holds.}$$

## 9 Uniqueness for the Cauchy Problem

In Sec. 7 we proved the existence of a solution to the Cauchy problem

(7.1), (7.2) satisfying (7.7). We now prove a uniqueness theorem.

**Theorem 16** Let  $L$  satisfy the assumptions  $(A_i)'$ ,  $(A_3)'$  in  $O = \mathbb{R}^n \times X$ .

$[0, 2']$ . Then there exists at most one solution to the Cauchy problem  
 $\hookrightarrow$  (7.1),  
 (7.2) satisfying the boundedness condition

$$(9L) \quad |L(a; W)| \leq C \exp[-\lambda |W|] \quad dx dt \ll \langle \rangle$$

is  $h \gg$

for some positive number  $k$ .

It follows that under the assumptions of Theorem 12, Sec. 7, and the

additional assumption  $(A_3)'$ , there exists a unique solution to the Cauchy problem (7.1), (7.2) satisfying (7.7).

*Proof.* We have to prove that if  $Lu(x, t) = 0$ ,  $u(x, 0) = 0$  then  $u(x, t) =$

0. We shall first prove that  $u(x, t) = 0$  if  $0 < t < d$  for some  $d$  sufficiently small. Let  $(x, t)$  be an arbitrary fixed point with  $0 < t < d$ . We have to prove that  $m(kh, t) = 0$ . Denote by  $B_r$  a ball in  $\mathbb{R}^n$  with center  $x_0$  and radius

$R$  and let  $B_r^c$  be the complement of  $B_r$  in  $V_{t+s} \setminus$ .

There exists a function  $L(PS)$  twice continuously  
 $\hookrightarrow$  differentiable in  $\mathbb{R}^n$

and having the following properties (see Problem 4): 1 if  $PS \in B_r$ ,

$$(9.2) \quad L(EUR) =$$

o if  $PS \in B_r^c$   
 $dh(Q$

$$(9.3) \quad 0 < L(PS) < 1, \quad + S \\ \hookrightarrow < // u \in b''$$

$> i$  thx, thPS/ where  $I_a$  is a constant independent of  $R$ .

$$= \langle \langle \text{PS}, t \rangle, i \rangle = I_i(x)G(kh, t; x, g)$$

Integrating Green's identity (8.4) with  $i$

over the region  $\text{PS} \setminus B_{r+i}$ ,  $0 \leq i \leq s$  where  $s > J$ , and using  $w(a, 0) = 0$  we obtain, as  $\epsilon \rightarrow 0$ ,

$$(9.4) \quad \lim_{\epsilon \rightarrow 0} \int_{\text{PS} \setminus B_{r+i}} h(t) T(x, t; x, t) \langle \langle \text{PS}, t \rangle, i \rangle = \int_{\text{PS}} G$$

$$\int_{\text{PS}} u L^* \circ df \, dr;$$

$$\int_{\text{PS}} u L^* \circ df \, dr = \int_{\text{PS}} J \circ J \circ I \, dr$$

the integrals over

the boundary of  $B_{r+i}$  do not appear since  $I(\text{PS}) = 0$  outside  $B_{r+i}$ . The left-hand side of (9.4) is equal to  $h(x)u(x, I) = u(x, t)$ .

$= 0$  (which follows

Using (9.2) and  $L^*T(x, i; \text{PS}, t)$  by (8.13)) we find, that  $L^*v = 0$  if  $\text{PS} \neq \emptyset$ . Hence

$$(9.5) \quad \int_{\text{PS}} h(t) T(x, t; x, t) \langle \langle \text{PS}, t \rangle, i \rangle = \int_{\text{PS}} G$$

Now, from (8.15) and (8.13), (8.14) it follows that

$$|G(kh, i; t)| \leq C \int_{\text{PS}} G(1, \langle \langle \text{PS}, t \rangle, i \rangle, g)$$

$$\langle \langle \text{PS}, t \rangle, i \rangle = C \int_{\text{PS}} G(1, \langle \langle \text{PS}, t \rangle, i \rangle, g)$$

Substituting this into (9.5) we get

$$(9.6) \quad \int_{\text{PS}} u L^* \circ df \, dr \leq \text{const} \cdot \exp$$

We shall now make use of the condition (9.1). It implies that  
 $\tau < T_0^*$

$$K^* \int_0^{\tau} \exp[-2A(\tau-s)] \frac{1}{s} ds \leq \frac{1}{\tau} \int_0^{\tau} \exp[-2A(\tau-s)] ds$$

Hence,  
 $\tau \leq \frac{1}{K^*}$

$$(9.7) \quad \lim_{R \rightarrow \infty} \int_0^R \int_0^R |u(x, y)|^2 \exp[-2k(x+y)] dx dy = 0$$

by the

Comparing the integral of (9.7) with the right-hand side of (9.6) we see that if  $d < 10/8/c$  then the right-hand side of (9.6) also tends to zero as

$\tau \rightarrow 0$  if  $k \in G$

Rec. Thus,  $u(x, t) = 0$  for  $0 < t < d$  and  $d = 10/9/r$ . We can now proceed in the same way to prove that  $u(x, t) = 0$  in the strips  $d < t < 2d$ ,  $2d < t < 3d$ , etc., and the proof is thereby completed.

We conclude

this section by giving an example of a solution  $u(x, t) \geq 0$  of the heat equation (1.1)

$$u_{xx} - u_t = 0 \quad (-\infty < x < \infty, \quad 0 < t < \infty)$$

satisfying the initial condition

$$u(x, 0) = 0 \quad (|x| \leq d)$$

and the boundedness condition

$$(9.8) \quad \frac{dx}{dt} \leq \frac{d}{dt} \int_{-\infty}^{\infty} |u(x, t)|^2 dx$$

where  $\epsilon$  is any given positive number.

It is known (see [80]) that for any  $d > 0$  there exist infinitely differentiable functions  $f(t)$  such that  $f(t) \geq 0$  and  $f(t) = 0$  for  $t \leq d$  satisfying the conditions: (PS)

0 if  $i < u$  and if  $i > 1$  and =

$$|f| < C z^{\alpha} e^{\beta} \quad (m = 1, 2, \dots)$$

for  $0 < t < 1$ , where  $S$  is a constant. Now take

$$(9.9) \quad \dots = S \dots a^*.$$

$t_i U, i) J t^{\wedge}$

The series and its first two derivatives are uniformly convergent in bounded

sets of  $\{x, t\}$ , provided  $d < 1$ . Since

$$E^*, \quad 0 < L + S \\ w^{2l+6} w \\ < S, \quad e_{kh} \leq 4|kh|'$$

where the  $S_g$ - are constants and  $17 = 2/(1 - d)$ , we can take  $d$  sufficiently small so that  $e < 2 + e$  and, consequently, (9.8) is satisfied.

The above example shows that Theorem 1S is not true if in the condition -

(9.1) we replace  $\exp[-\psi|a|/2]$  by  $\exp[-\psi|a|/2 + \epsilon]$ ,  $\epsilon > 0$ .

## 10 Problems

1. Let  $L, V$  be disjoint closed sets in  $\mathbb{R}^n$  and let  $L$  be bounded. Prove that there

exists an infinitely differentiable function  $f$  in  $\mathbb{R}^n$  such that  $f(x) = 1$  if  $x \in L$ ,  $f(x) = 0$  if  $x \in V$ , and  $0 < f(x) < 1$  for all  $x \notin L \cup V$ .

[Hint: Let  $d = \text{dist}(L, V)$  and denote by  $\Omega_d$  the set of all points whose distance to  $L$  is  $< d/2$ . Take

$$h(x) = \int_{\Omega_d} p(x-y) dy \quad (e = d/3)$$

$U \cap \Omega_d$  where

$$G_0 \quad \text{if } |x| > e,$$

$r_L(i)$

$$\text{if } W < e -$$

if

$$1^{-E}[-E=p<<]$$

2. Prove Theorem 14. Sec. S.

Let  $f$  be any function on  $\mathbb{R}^n$  which is infinitely differentiable and has compact support in  $\Omega$ , satisfying  $\int_{\Omega} f(x) dx = 0$ . Use the relation  $\int_{\Omega} (L^* f - f L) dx = 0$ .

3. For any linear differential operator with sufficiently smooth coefficients in a domain  $\Omega$ , we define its adjoint by

$$L^* u = \sum_{|\alpha| \leq m} (-1)^{|\alpha|} \frac{\partial}{\partial x} \left( a_{\alpha} \frac{\partial u}{\partial x} \right) + \sum_{|\alpha| \leq m-1} b_{\alpha} \frac{\partial u}{\partial x} + c u$$
 If  $L^* = L$  then we say that  $L$  is self-adjoint. Prove that this definition of adjoint

is equivalent to the following one:  $L^*$  is a linear differential operator for which

$$\int_{\Omega} (v L u - u L^* v) dx = 0$$

for all infinitely differentiable functions  $u, v$  having a compact support in  $\Omega$ .

4. Prove that there exists a function  $L(x, y)$  satisfying (0.2), (0.3).  
[Hint: Use the  $L$  in Problem 1 with  $A_0 = \frac{1}{2}$ ,  $e = 1/4$ .]  
5. Let  $G(x, y; x, t)$  be the fundamental solution of  $L u = 0$  in  $B_n \times [T_0, \infty)$  (under the assumptions  $(A_1)$ ,  $(A_2)$ ). Prove:

$$T(x, t; x, t) = \int_{B_n} G(x, y; x, t) T(y, s; x, r) dy, \quad (r < s < t).$$

Proof

6. Show that the assumption  $h < 10/4G$  made in Theorem 12 cannot be relaxed.  
[Hint: Take  $u(x, t) = \exp[-4At] \exp[-(1-4Li)]$ .]  
7. Verify that all the results of Chap. I remain true if the conditions (1.5), (1.6)

are replaced by the following weaker conditions: For every compact subset  $K$

of  $D$  there exists a constant  $A_K$  depending on  $K$ , such that

$$|b_i(x, t) - Y(x, t, 1)| \leq A_K |x - x'|$$

--

$$|c(x,t) - c(x_{deg},f)| < A' \|x - x_{deg}\|^a,$$

for all  $x \in K$ ,  $x_{deg} \in K$ ,  $T_0 < t < T_1$ . Verify also that the assumption that  $b, s$  are continuous functions throughout  $O$  may be replaced by the assumption that

$b, s$  are bounded continuous functions in  $D \cup K_h \cup (G_0, G_1]$ .